

# Clima App

## Reporte Técnico del Proyecto

|                 |                                    |
|-----------------|------------------------------------|
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| <b>Materia:</b> | Práctica 2.2                       |
| <b>Tipo:</b>    | Aplicación Flutter Multiplataforma |
| <b>Versión:</b> | 1.0.0+1                            |
| <b>Fecha:</b>   | Junio 2026                         |

*Generado automáticamente con fpdf2 (Python)*

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# 1. Resumen del Proyecto

Clima App es una aplicación móvil multiplataforma desarrollada con Flutter como parte de la práctica 2.2 de la materia. La aplicación muestra información meteorológica simulada (datos hardcodeados) para tres ciudades: Santiago de Querétaro, Querétaro y México. Incluye funcionalidades de búsqueda de ciudades, visualización del clima actual y un pronóstico extendido a 5 días.

## Información General

|                          |                                          |
|--------------------------|------------------------------------------|
| <b>Nombre:</b>           | Clima App                                |
| <b>Paquete:</b>          | com.example.clima_app                    |
| <b>Versión:</b>          | 1.0.0+1                                  |
| <b>SDK Dart:</b>         | ^3.12.0                                  |
| <b>Plataformas:</b>      | Android, iOS, Web, Windows, macOS, Linux |
| <b>Lenguaje:</b>         | Dart (null safety)                       |
| <b>UI Framework:</b>     | Material Design 3                        |
| <b>Archivos Dart:</b>    | 7 (lib/), 1 test                         |
| <b>Líneas de código:</b> | ~420 Dart                                |

## Funcionalidades Principales

- Visualización del clima actual con temperatura, humedad y viento
- Búsqueda y filtrado de ciudades en tiempo real
- Pronóstico meteorológico a 5 días
- Diseño responsivo (orientación portrait y landscape)
- Interfaz con Material Design 3
- Iconos climáticos según la condición (soleado, nublado, lluvioso)

## 2. Tecnologías y Framework

| Tecnología | Detalle                      | Versión          |
|------------|------------------------------|------------------|
| Flutter    | Framework UI multiplataforma | Stable (3.x)     |
| Dart       | Lenguaje de programación     | ^3.12.0          |
| Material 3 | Diseño de interfaz           | Última           |
| Android    | Plataforma móvil             | AGP 9.0.1        |
| Kotlin     | Android nativo               | 2.3.20           |
| Java       | Compatibilidad Android       | 17               |
| iOS        | Plataforma móvil             | Swift/Storyboard |
| Web        | PWA despliegue web           | HTML5            |

### Herramientas de Desarrollo

- Editor: IntelliJ IDEA / Android Studio
- Control de versiones: Git
- Linter: flutter\_lints ^6.0.0
- Testing: flutter\_test (widget testing)
- Gradle: Wrapper con AGP 9.0.1

### 3. Estructura del Proyecto

El proyecto sigue la estructura estándar de Flutter. El código fuente principal se encuentra en el directorio lib/, organizado en las subcarpetas screens/ y widgets/.

```
clima_app/
+-- lib/
|   +-- main.dart           # Punto de entrada
|   +-- screens/
|       +-- home_screen.dart # Pantalla principal
|       +-- search_screen.dart # Búsqueda de ciudades
|       +-- detail_screen.dart # Pronóstico 5 días
|   +-- widgets/
|       +-- custom_button.dart # Boton personalizado
|       +-- weather_icon.dart  # Icono de clima
|       +-- temperature_card.dart # Tarjeta de temperatura
+-- test/
|   +-- widget_test.dart      # Prueba basica
+-- pubspec.yaml              # Configuración y dependencias
+-- android/                  # Configuración Android
+-- ios/                       # Configuración iOS
+-- web/                       # Configuración Web
+-- windows/                   # Configuración Windows
+-- macos/                     # Configuración macOS
+-- linux/                     # Configuración Linux
```

#### lib/main.dart

Punto de entrada de la aplicación. Define MyApp como StatelessWidget con MaterialApp, tema azul y Material 3. Ruta inicial: HomeScreen.

```
import 'package:flutter/material.dart';
import 'screens/home_screen.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Climate App',
      theme: ThemeData(
        primarySwatch: Colors.blue,
        useMaterial3: true,
      ),
      home: const HomeScreen(),
    );
  }
}
```

## lib\screens/home\_screen.dart

Pantalla principal con datos climáticos hardcodedos (24°C, humedad 65%, viento 12 km/h). Icono nublado. Diseño responsivo portrait/landscape. Navega a SearchScreen.

```
import 'package:flutter/material.dart';
import '../widgets/weather_icon.dart';
import '../widgets/custom_button.dart';
import 'search_screen.dart';

class HomeScreen extends StatelessWidget {
  const HomeScreen({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) { // ? 'context' está definido AQUÍ
    final isLandscape = MediaQuery.of(context).orientation == Orientation.landscape;

    return Scaffold(
      appBar: AppBar(
        title: const Text('Clima Actual'),
        centerTitle: true,
      ),
      body: SafeArea(
        child: SingleChildScrollView(
          padding: const EdgeInsets.all(16.0),
          child: isLandscape
            ? _buildLandscapeLayout(context) // ? PASAS context como parámetro
            : _buildPortraitLayout(context), // ? PASAS context como parámetro
        ),
      ),
    );
  }

  // Layout vertical - recibe context como parámetro
  Widget _buildPortraitLayout(BuildContext context) {
    return Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        const Text(
          '24°C',
          style: TextStyle(
            fontSize: 72,
            fontWeight: FontWeight.bold,
            color: Colors.blue,
          ),
        ),
        const SizedBox(height: 16),
        const Text(
          'Santiago de Querétaro',
          style: TextStyle(fontSize: 24),
        ),
        const SizedBox(height: 32),
        const WeatherIcon(condition: 'cloudy', size: 120),
        const SizedBox(height: 32),
        const Text(
          'Humedad: 65% | Viento: 12 km/h',
          style: TextStyle(fontSize: 16),
        ),
        const SizedBox(height: 40),
        CustomButton(
          text: 'Buscar Ciudades',
        ),
      ],
    );
  }
}
```

```
        icon: Icons.search,
        onPressed: () {
          Navigator.push(
            context, // ? Ahora context existe porque lo pasamos
            MaterialPageRoute(builder: (context) => const SearchScreen()),
          );
        },
      ),
      const SizedBox(height: 20),
    ],
  );
}
```

// Layout horizontal - recibe context como parámetro

```
Widget _buildLandscapeLayout(BuildContext context) {
  return Row(
    crossAxisAlignment: CrossAxisAlignment.center,
    mainAxisAlignment: MainAxisAlignment.spaceEvenly,
    children: [
      // Columna izquierda
      Expanded(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: const [
            Text(
              '24°C',
              style: TextStyle(
                fontSize: 60,
                fontWeight: FontWeight.bold,
                color: Colors.blue,
              ),
              textAlign: TextAlign.center,
            ),
            SizedBox(height: 12),
            Text(
              'Santiago de Querétaro',
              style: TextStyle(fontSize: 18),
              textAlign: TextAlign.center,
            ),
          ],
        ),
      ),
      // Columna derecha
      Expanded(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            const WeatherIcon(condition: 'cloudy', size: 80),
            const SizedBox(height: 16),
            const Text(
              'Humedad: 65% | Viento: 12 km/h',
              style: TextStyle(fontSize: 14),
              textAlign: TextAlign.center,
            ),
            const SizedBox(height: 24),
            CustomButton(
```



```
      text: 'Buscar Ciudades',
      icon: Icons.search,
      onPressed: () {
        Navigator.push(
          context, // ? Ahora context existe
          MaterialPageRoute(builder: (context) => const SearchScreen()),
        );
      },
    ),
  ],
),
],
);
}
```

## lib\screens/search\_screen.dart

Pantalla de búsqueda con lista hardcodeada de 3 ciudades. TextField con filtro en tiempo real. Navega a DetailScreen.

```
import 'package:flutter/material.dart';
import 'detail_screen.dart';

class SearchScreen extends StatefulWidget {
  const SearchScreen({Key? key}) : super(key: key);

  @override
  State<SearchScreen> createState() => _SearchScreenState();
}

class _SearchScreenState extends State<SearchScreen> {
  // Lista de ciudades con sus temperaturas simuladas
  final List<Map<String, String>> cities = [
    {'name': 'Santiago', 'temp': '24°C'},
    {'name': 'Querétaro', 'temp': '22°C'},
    {'name': 'México', 'temp': '20°C'},
  ];

  List<Map<String, String>> filteredCities = [];

  @override
  void initState() {
    super.initState();
    filteredCities = cities;
  }

  void filterCities(String query) {
    setState(() {
      if (query.isEmpty) {
        filteredCities = cities;
      } else {
        filteredCities = cities
          .where((city) =>
            city['name']!.toLowerCase().contains(query.toLowerCase()))
          .toList();
      }
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Buscar Ciudades'),
      ),
      body: Column(
        children: [
          Padding(
            padding: const EdgeInsets.all(16.0),
            child: TextField(
              onChanged: filterCities,
              decoration: const InputDecoration(
                hintText: 'Busca una ciudad...',
                prefixIcon: Icon(Icons.search),
                border: OutlineInputBorder(),
              ),
            ),
          ),
        ],
      ),
    );
  }
}
```

```

        suffixIcon: Icon(Icons.clear),
      ),
    ),
  ),
  Expanded(
    child: filteredCities.isEmpty
      ? const Center(
          child: Text('No se encontraron ciudades'),
        )
      : ListView.builder(
          itemCount: filteredCities.length,
          itemBuilder: (context, index) {
            final city = filteredCities[index];
            return Card(
              margin: const EdgeInsets.symmetric(
                horizontal: 16, vertical: 4),
              child: ListTile(
                leading: const Icon(Icons.location_city),
                title: Text(
                  city['name']!,
                  style: const TextStyle(fontWeight: FontWeight.bold),
                ),
                subtitle: Text('Temperatura: ${city['temp']}'),
                trailing: const Icon(Icons.chevron_right),
                onTap: () {
                  Navigator.push(
                    context,
                    MaterialPageRoute(
                      builder: (context) =>
                        DetailScreen(city: city['name']!),
                    ),
                  );
                },
              ),
            );
          },
        ),
      ),
    ),
  ],
),
);
}
}

```

## lib/screens/detail\_screen.dart

Pronóstico de 5 días (lunes a viernes) con temperaturas y emojis climáticos. Botón Volver.

```
import 'package:flutter/material.dart';
class DetailScreen extends StatelessWidget {
  final String city;
  const DetailScreen({Key? key, required this.city}) : super(key: key);
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('$city - 5 Días')),
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
          children: [
            Row(
              mainAxisAlignment: MainAxisAlignment.spaceEvenly,
              children: const [
                Text('Lun\n24°C\n? '),
                Text('Mar\n26°C\n??'),
                Text('Mié\n20°C\n??'),
                Text('Jue\n25°C\n??'),
                Text('Vie\n28°C\n??'),
              ],
            ),
            const SizedBox(height: 40),
            ElevatedButton(
              onPressed: () => Navigator.pop(context),
              child: const Text('Volver'),
            ),
          ],
        ),
      );
  }
}
```

## lib\widgets/custom\_button.dart

Botón reutilizable basado en ElevatedButton.icon. Configurable: texto, icono, color, onPressed.

```
import 'package:flutter/material.dart';

class CustomButton extends StatelessWidget {
  final String text;
  final VoidCallback onPressed;
  final IconData? icon;
  final Color? color;

  const CustomButton({
    Key? key,
    required this.text,
    required this.onPressed,
    this.icon,
    this.color,
  }) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return ElevatedButton.icon(
      onPressed: onPressed,
      icon: icon != null ? Icon(icon) : const SizedBox.shrink(),
      label: Text(text),
      style: ElevatedButton.styleFrom(
        backgroundColor: color ?? Colors.blue,
        foregroundColor: Colors.white,
        padding: const EdgeInsets.symmetric(horizontal: 24, vertical: 12),
        shape: RoundedRectangleBorder(
          borderRadius: BorderRadius.circular(8),
        ),
      ),
    );
  }
}
```

## lib\widgets/weather\_icon.dart

Mapea condiciones climáticas a iconos Material: sunny, cloudy, rainy. Tamaño y color configurables.

```
import 'package:flutter/material.dart';

class WeatherIcon extends StatelessWidget {
  final String condition;
  final double size;
  final Color? color;

  const WeatherIcon({
    Key? key,
    required this.condition,
    this.size = 80,
    this.color,
  }) : super(key: key);

  @override
  Widget build(BuildContext context) {
    IconData iconData;

    switch (condition.toLowerCase()) {
      case 'sunny':
        iconData = Icons.wb_sunny;
        break;
      case 'cloudy':
        iconData = Icons.cloud;
        break;
      case 'rainy':
        iconData = Icons.umbrella;
        break;
      default:
        iconData = Icons.cloud_queue;
    }

    return Icon(
      iconData,
      size: size,
      color: color ?? Colors.blue,
    );
  }
}
```

## lib\widgets/temperature\_card.dart

Tarjeta con gradiente azul mostrando temperatura y ciudad. Soporta onTap. Esquinas redondeadas.

```
import 'package:flutter/material.dart';

class TemperatureCard extends StatelessWidget {
  final String temperature;
  final String city;
  final VoidCallback? onTap;

  const TemperatureCard({
    Key? key,
    required this.temperature,
    required this.city,
    this.onTap,
  }) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return Card(
      elevation: 4,
      shape: RoundedRectangleBorder(
        borderRadius: BorderRadius.circular(12),
      ),
      child: InkWell(
        onTap: onTap,
        borderRadius: BorderRadius.circular(12),
        child: Container(
          padding: const EdgeInsets.all(16),
          decoration: BoxDecoration(
            borderRadius: BorderRadius.circular(12),
            gradient: const LinearGradient(
              colors: [Colors.blue, Colors.blueAccent],
              begin: Alignment.topLeft,
              end: Alignment.bottomRight,
            ),
          ),
          child: Column(
            mainAxisAlignment: MainAxisAlignment.min,
            children: [
              Text(
                temperature,
                style: const TextStyle(
                  fontSize: 28,
                  fontWeight: FontWeight.bold,
                  color: Colors.white,
                ),
              ),
              const SizedBox(height: 8),
              Text(
                city,
                style: const TextStyle(
                  fontSize: 16,
                  color: Colors.white,
                ),
              ),
            ],
          ),
        ),
      ),
    );
  }
}
```

```
    ),  
    );  
}  
}
```

**pubspec.yaml**



```
name: clima_app
description: "A new Flutter project."
# The following line prevents the package from being accidentally published to
# pub.dev using `flutter pub publish`. This is preferred for private packages.
publish_to: 'none' # Remove this line if you wish to publish to pub.dev

# The following defines the version and build number for your application.
# A version number is three numbers separated by dots, like 1.2.43
# followed by an optional build number separated by a +.
# Both the version and the builder number may be overridden in flutter
# build by specifying --build-name and --build-number, respectively.
# In Android, build-name is used as versionName while build-number used as versionCode.
# Read more about Android versioning at https://developer.android.com/studio/publish/versioning
# In iOS, build-name is used as CFBundleShortVersionString while build-number is used as CFBundleVersion.
# Read more about iOS versioning at
# https://developer.apple.com/library/archive/documentation/General/Reference/InfoPlistKeyReference/Articles/CoreFoundationKeys
# In Windows, build-name is used as the major, minor, and patch parts
# of the product and file versions while build-number is used as the build suffix.
version: 1.0.0+1

environment:
  sdk: ^3.12.0

# Dependencies specify other packages that your package needs in order to work.
# To automatically upgrade your package dependencies to the latest versions
# consider running `flutter pub upgrade --major-versions`. Alternatively,
# dependencies can be manually updated by changing the version numbers below to
# the latest version available on pub.dev. To see which dependencies have newer
# versions available, run `flutter pub outdated`.
dependencies:
  flutter:
    sdk: flutter

  # The following adds the Cupertino Icons font to your application.
  # Use with the CupertinoIcons class for iOS style icons.
  cupertino_icons: ^1.0.8

dev_dependencies:
  flutter_test:
    sdk: flutter

  # The "flutter_lints" package below contains a set of recommended lints to
  # encourage good coding practices. The lint set provided by the package is
  # activated in the `analysis_options.yaml` file located at the root of your
  # package. See that file for information about deactivating specific lint
  # rules and activating additional ones.
  flutter_lints: ^6.0.0

# For information on the generic Dart part of this file, see the
# following page: https://dart.dev/tools/pub/pubspec

# The following section is specific to Flutter packages.
flutter:

  # The following line ensures that the Material Icons font is
  # included with your application, so that you can use the icons in
```

```
# the material Icons class.
uses-material-design: true

# To add assets to your application, add an assets section, like this:
# assets:
#   - images/a_dot_burr.jpeg
#   - images/a_dot_ham.jpeg

# An image asset can refer to one or more resolution-specific "variants", see
# https://flutter.dev/to/resolution-aware-images

# For details regarding adding assets from package dependencies, see
# https://flutter.dev/to/asset-from-package

# To add custom fonts to your application, add a fonts section here,
# in this "flutter" section. Each entry in this list should have a
# "family" key with the font family name, and a "fonts" key with a
# list giving the asset and other descriptors for the font. For
# example:
# fonts:
#   - family: Schyler
#     fonts:
#       - asset: fonts/Schyler-Regular.ttf
#       - asset: fonts/Schyler-Italic.ttf
#         style: italic
#   - family: Trajan Pro
#     fonts:
#       - asset: fonts/TrajanPro.ttf
#       - asset: fonts/TrajanPro_Bold.ttf
#         weight: 700
#
# For details regarding fonts from package dependencies,
# see https://flutter.dev/to/font-from-package
```

## 5. Dependencias

| Dependencia     | Versión | Tipo    |
|-----------------|---------|---------|
| flutter         | SDK     | Directa |
| cupertino_icons | ^1.0.8  | Directa |
| flutter_test    | SDK     | Dev     |
| flutter_lints   | ^6.0.0  | Dev     |

El proyecto tiene dependencias mínimas. No utiliza paquetes externos para consumo de APIs, ya que todos los datos meteorológicos están hardcodeados. Esto es consistente con un proyecto académico enfocado en la estructura de UI y navegación.

## 6. Historial de Git

| Commit  | Fecha      | Mensaje                                |
|---------|------------|----------------------------------------|
| 3543d3f | 2026-06-01 | Práctica 2.2 (+5381 líneas)            |
| e3e771b | 2026-05-29 | Práctica 2.1 Hot Reload                |
| 9816b97 | 2026-05-29 | Práctica 1.3 Wearable y Smart TV       |
| 0ac6021 | 2026-05-28 | Creación de componentes práctica 1.2   |
| 89a6a42 | 2026-05-26 | Diseño en Figma de aplicación de clima |
| 1026a60 | 2026-05-26 | Cambio de nombre de práctica           |
| ac5f3c4 | 2026-05-23 | Segunda práctica (init flutter)        |
| 5a3a486 | 2026-05-20 | Práctica 1.1                           |

## Evolución del Proyecto

- P1.1 (20-may): Documento inicial de la práctica
- P1.2 (23-may): Creación del proyecto Flutter (Hello World)
- P1.3 (29-may): Estudio de adaptación Wearable/Smart TV
- P2.1 (29-may): Modificaciones con Hot Reload
- P2.2 (01-jun): Refactor mayor a estructura screens/widgets

## 7. Estadísticas del Proyecto

| Métrica                          | Valor        |
|----------------------------------|--------------|
| Archivos Dart (lib/)             | 6            |
| Archivos de prueba               | 1            |
| Líneas de código Dart            | ~420         |
| Archivos totales (sin build/)    | ~850         |
| Líneas agregadas (último commit) | +5,381       |
| Plataformas objetivo             | 6            |
| Commits totales                  | 8            |
| Contribuyentes                   | 1 (ruben727) |

### Distribución de Código por Archivo

| Archivo                           | Líneas | %   |
|-----------------------------------|--------|-----|
| lib/main.dart                     | 22     | 5%  |
| lib/screens/home_screen.dart      | 128    | 31% |
| lib/screens/search_screen.dart    | 99     | 24% |
| lib/screens/detail_screen.dart    | 33     | 8%  |
| lib/widgets/custom_button.dart    | 33     | 8%  |
| lib/widgets/weather_icon.dart     | 39     | 9%  |
| lib/widgets/temperature_card.dart | 60     | 14% |

## 8. Capturas de Pantalla

No se encontraron capturas de pantalla en el proyecto. Las imágenes de diseño (Figma) se encuentran en commits anteriores.